

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected December 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C/C++, Python, Assembly, Java

Embedded Firmware: STM32H7, TTC 60 VCU, PX4, real-time control, watchdogs, microcontroller I/O

Protocols & Linux: CAN, SPI, Linux SocketCAN, python-can, WebSockets

Test & Debug: Board bring-up, software-in-the-loop, scripted replay, GDB, BusMaster, TTC Downloader, telemetry traces

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Leading an in-house VCU program replacing a vendor controller with custom STM32H755 hardware, running a **10 ms** deterministic control task under a watchdog that reboots the controller if the task hangs
- Built its dual-bus CAN driver and identified non-**5 V**-tolerant analog inputs during bring-up, specifying voltage dividers before vehicle integration
- Developing the custom-inverter path for a **four-motor** EV, mapping calibrated pedal input to bounded CAN current commands with fail-safe precharge and ready-to-drive gating
- Reworked custom-BMS integration, decoding **46 CAN message IDs** across **96 cells**, **96 thermistors**, and **8 modules**, with a **1 s** communication timeout

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated embedded flight-control firmware in C on PX4, using structured telemetry logs and repeatable flight tests to reproduce issues and evaluate revisions
- Integrated and tuned visual-inertial odometry for onboard state estimation, comparing configuration changes against logged position-hold behavior

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Developed fixed-point PI firmware with saturation, anti-windup, and slew limiting that held measured power within **0.01%** of its commanded setpoint while enforcing an **80 kW** limit
- Built CAN diagnostics firmware for live power and torque frames, status bits, and error counters for real-time BusMaster monitoring and regression review

PROJECT EXPERIENCE

STM32 Driver Display, Embedded Instrumentation

Feb 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

C, STM32H7, CAN, SPI

- Developed STM32H7 firmware that receives real-time CAN telemetry and drives an SPI LCD plus LED alerts for high-voltage, throttle, temperature, and power-budget status

SR-Wireless-CAN, Linux Firmware Deployment & Test Tool

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, Linux SocketCAN, python-can, FastAPI, WebSockets, SQLite, Docker

- Reverse-engineered and implemented an **8-stage** CAN firmware-deployment workflow from vendor traces, validating acknowledgements and CRC results over Linux SocketCAN
- Added simulated, recorded-trace, and live-CAN modes and documented frame encoding, timeouts, configuration, and recovery steps for repeatable debugging